

Question number	Answer	Mark
7(a)(i)	The only correct answer is C. <i>A is incorrect because DNA polymerase is used to synthesis DNA</i> <i>B is incorrect because integrase inserts DNA into other DNA</i> <i>D is incorrect because reverse transcriptase synthesises DNA using a RNA template</i>	(1)

Question number	Answer	Additional guidance	Mark
7(a)(ii)	An explanation that includes three the following points: • because restriction enzymes recognise specific (base) sequences (1) • because these recognition sites are not equally spaced along the DNA (1) • the enzyme hydrolyses the phosphodiester bonds (1) • therefore different sized fragments move different distances through the gel (1)	IGNORE incorrectly named enzyme from part (i) ACCEPT different enzymes cut at different sites ACCEPT different cutting sites will result in different length fragments ACCEPT breaks ACCEPT different speeds	(3)

Question number	Answer	Additional guidance	Mark
7(b)(i)	0.2 (1) 794 / 871 / 873 / 1000 (1)		(2)

Question number	Answer	Additional guidance	Mark
7(b)(ii)	indication of spot positioned between the origin and the 10 000 spot	DO NOT ACCEPT if clearly overlapping CE from 7(b)(i) i.e. if answer in 7(b)(i) is greater than 0.25, the spot would be below the 10 000 spot	(1)

Question number	Answer	Additional guidance	Mark
7(c)	{more molecules / more bonds / smaller spaces / fewer spaces} so higher {resistance / friction}	ACCEPT harder to move for higher resistance	(1)

Question number	Answer	Additional guidance	Mark
7(d)(i)	An answer that includes two of the following points: • bacterial {genome / chromosome} / (bacterial) nucleoid (1) • plasmid (1) • in mitochondria / mitochondrial DNA / mtDNA (1) • in chloroplasts / chloroplast DNA / cpDNA (1)	IGNORE loops of DNA in bacteria IGNORE mitochondria / chloroplasts unqualified ACCEPT cccDNA (formed by some viruses inside cell nuclei) viral DNA inserted into cell {ecc / extrachromosomal circular} DNA DO NOT ACCEPT ribosomes / in cytoplasm / in viruses	(2)

Question number	Answer	Additional guidance	Mark
7(d)(ii)	An answer that includes two of the following points: • linear DNA has (3' and 5') ends but circular DNA does not (1) • linear DNA is associated with (more) {proteins / histones} than circular DNA (1) • linear DNA will have (unbound) {phosphate / deoxyribose} but circular DNA will not (1) • linear DNA will have (one) fewer phosphodiester bond than circular DNA (with the same number of mononucleotides) (1)	ACCEPT converse ACCEPT circular DNA is supercoiled but linear DNA is not ACCEPT converse	(2)